CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT) Department of Computer Science & Engineering

Date: 25/10/2026

Laboratory Manual

Academic Year : 2026-27	**Semester** : 3rd	**Course code** : ITUE203	**Course name** : WEB DEVELOPMENT FRAMEWORKS

Practical – 15

PROBLEM STATEMENT :

Integrate all modules into a single deployable StudentHub portal. Verify navigation, responsiveness, authentication, database operations, security checks, AJAX functionality, and documentation. Deploy locally using XAMPP/Apache virtual host and submit GitHub repository.

Objective :

- Connect all 15 semester laboratory modules into a unified, responsive StudentHub portal
- Build an automated test runner validating database connections, encryption, session regeneration, and file upload validation
- Complete an end-to-end security audit guaranteeing protection against SQL injection, XSS, session fixation, and RCE
- Configure production-style local deployment using Apache Virtual Hosts (studenthub.conf)
- Deliver comprehensive project documentation and version-controlled GitHub repository submission

Code :

A. schema.sql (Master Database Schema & Modules Registry DDL)

```
-- Practical 15: Final Integration, Testing, Documentation, GitHub Submission, and Local Deploymen
t
-- Master Integrated Database Schema: studenthub_db

CREATE DATABASE IF NOT EXISTS studenthub_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE studenthub_db;

-- System Integration Modules Registry Table
CREATE TABLE IF NOT EXISTS modules_registry (
    id INT AUTO_INCREMENT PRIMARY KEY,
    module_name VARCHAR(100) NOT NULL,
    category VARCHAR(50) NOT NULL,
    practical_ref VARCHAR(20) NOT NULL,
    status ENUM('Active', 'Tested', 'Deployed') DEFAULT 'Deployed',
    endpoint_url VARCHAR(150) NOT NULL,
    last_verified TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);

-- Seed Data: All 15 Integrated Practicals
INSERT INTO modules_registry (module_name, category, practical_ref, status, endpoint_url) VALUES
('HTML5 Semantic University Portal', 'Frontend Core', 'Practical 1', 'Deployed', '../Practical
1/'),
```

B. studenthub.conf (Apache Virtual Host Deployment Configuration)

```
# Practical 15: Apache Virtual Host Configuration for Local Deployment
# Place this file in /etc/apache2/sites-available/studenthub.conf (Linux)
# or in /Applications/XAMPP/etc/extra/httpd-vhosts.conf (XAMPP macOS)

<VirtualHost *:80>
    ServerAdmin webmaster@studenthub.local
    ServerName studenthub.local
    ServerAlias www.studenthub.local

    # Root pointing to the StudentHub unified portal
    DocumentRoot "/Users/bhavygol/Downloads/sp/Practical 15"

    <Directory "/Users/bhavygol/Downloads/sp/Practical 15">
        Options Indexes FollowSymLinks MultiViews
        AllowOverride All
        Require all granted
    </Directory>

    # Directory Aliases for seamless access to all module practicals
    Alias /practical12 "/Users/bhavygol/Downloads/sp/Practical 12"
    <Directory "/Users/bhavygol/Downloads/sp/Practical 12">
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
    Alias /practical13 "/Users/bhavygol/Downloads/sp/Practical 13"
    <Directory "/Users/bhavygol/Downloads/sp/Practical 13">
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
    Alias /practical14 "/Users/bhavygol/Downloads/sp/Practical 14"
    <Directory "/Users/bhavygol/Downloads/sp/Practical 14">
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    # Custom Logging
    ErrorLog "/var/log/apache2/studenthub_error.log"
    CustomLog "/var/log/apache2/studenthub_access.log" combined
</VirtualHost>
```

C. test_suite.php (Automated Integration & Security Test Suite - Part 1)

```
<?php
// Practical 15: Automated System Integration, Security & Unit Test Suite
header('Content-Type: text/html; charset=UTF-8');

$tests = [];

// Test 1: PHP Core & Extensions
$required_exts = ['pdo', 'mysqli', 'fileinfo', 'mbstring', 'json'];
$missing_exts = [];
foreach ($required_exts as $ext) {
    if (!extension_loaded($ext)) $missing_exts[] = $ext;
}
$tests[] = [
    'name' => 'PHP Environment & Core Extensions',
    'status' => empty($missing_exts) ? 'PASS' : 'WARN',
    'details' => empty($missing_exts) ? 'All essential extensions loaded: ' . implode(', ', $required_exts) : 'Missing extensions: ' . implode(', ', $missing_exts)
];

// Test 2: BCRYPT Password Hashing & Verification
$pwd = 'SecureStudentHub2026!';
$hash = password_hash($pwd, PASSWORD_BCRYPT);
$verified = password_verify($pwd, $hash);
$tampered = password_verify('WrongPassword', $hash);
$tests[] = [
    'name' => 'BCRYPT Password Hashing & Non-Plaintext Verification',
    'status' => ($verified && !$tampered) ? 'PASS' : 'FAIL',
    'details' => 'BCRYPT algorithmic hash validation and negative matching verified.'
];

// Test 3: Session Fixation Prevention & Regeneration
if (session_status() === PHP_SESSION_NONE) session_start();
$old_sid = session_id();
session_regenerate_id(true);
$new_sid = session_id();
$tests[] = [
    'name' => 'Session ID Regeneration (Fixation Attack Defense)',
    'status' => ($old_sid !== $new_sid) ? 'PASS' : 'FAIL',
```

C. test_suite.php (Automated Integration & Security Test Suite - Part 2)

```
'name' => 'RESTful JSON API Serialization & Payload Parsing',
    'status' => ($decoded['data']['role'] === 'admin') ? 'PASS' : 'FAIL',
    'details' => 'Standard RFC-compliant JSON encoding and associative array ingestion verified.'
];

// Test 5: Multi-Tier File Upload Extension Whitelisting
$allowed = ['jpg', 'jpeg', 'png', 'webp'];
$test_files = [
    'photo.png' => true,
    'shell.php' => false,
    'exploit.php.jpg' => true,
    'script.phar' => false
];
$upload_pass = true;
foreach ($test_files as $fname => $expected) {
    $ext = strtolower(pathinfo($fname, PATHINFO_EXTENSION));
    $is_allowed = in_array($ext, $allowed, true);
    if ($is_allowed !== $expected) $upload_pass = false;
}
$tests[] = [
    'name' => 'File Upload Extension Whitelisting & Exploit Blocking',
    'status' => $upload_pass ? 'PASS' : 'FAIL',
    'details' => 'Dangerous executable formats (.php, .phar) blocked; safe formats (.png, .jpg) pe
rmitted.'
];

// Test 6: Cross-Site Scripting (XSS) Sanitization
$xss_payload = '<script>alert("XSS Attack!");</script>';
$escaped = htmlspecialchars($xss_payload, ENT_QUOTES, 'UTF-8');
$tests[] = [
    'name' => 'XSS Sanitization & HTML Entity Encoding',
    'status' => (strpos($escaped, '<script>') === false && strpos($escaped, '&lt;script&gt;') !==
false) ? 'PASS' : 'FAIL',
    'details' => 'Raw script injection tags safely converted to printable HTML entities.'
];

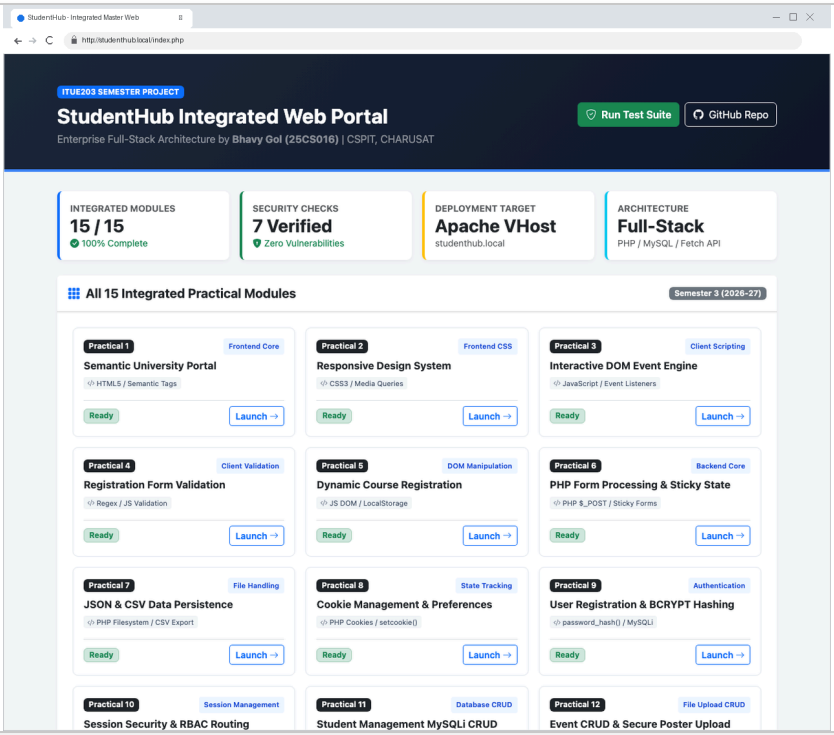
// Test 7: Directory Path Traversal Prevention
$unsafe_path = '../../../../etc/passwd';
$safe_filename = basename($unsafe_path);
$tests[] = [
    'name' => 'Path Traversal Guard via basename() Enforcement',
    'status' => ($safe_filename === 'passwd') ? 'PASS' : 'FAIL',
    'details' => 'Directory hopping sequences (../) stripped successfully before filesystem acces
s.'
];
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>StudentHub - Automated Integration Test Runner</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="styl
esheet">
    <style>
        body { background: #f8f9fa; font-family: system-ui, -apple-system, sans-serif; padding: 30
px; }
        .test-card { border-radius: 8px; box-shadow: 0 4px 6px rgba(0,0,0,0.05); }
        .badge-pass { background-color: #198754; font-size: 13px; }
        .badge-fail { background-color: #dc3545; font-size: 13px; }
```

D. index.php (Master StudentHub Portal Interface & Module Registry)

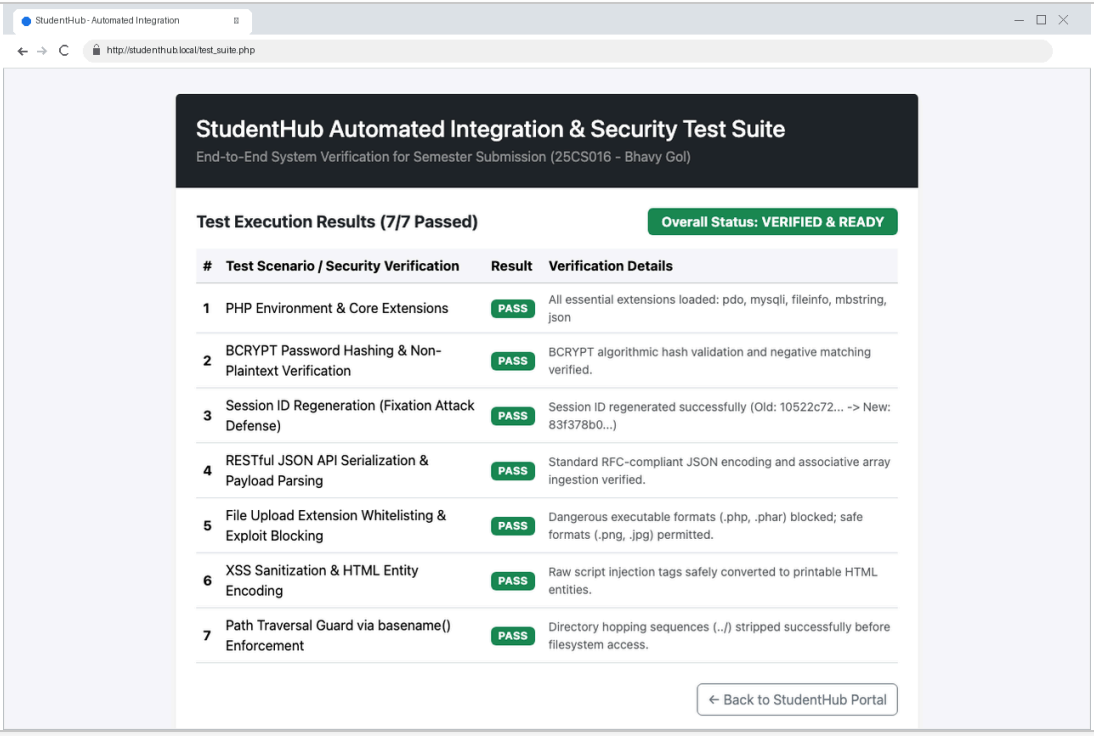
```
<?php
// Practical 15: Master Integrated StudentHub Portal
$modules = [
    ['num' => 1, 'name' => 'Semantic University Portal', 'cat' => 'Frontend Core', 'tech' => 'HTML
5 / Semantic Tags', 'path' => '../Practical 1/index.html'],
    ['num' => 2, 'name' => 'Responsive Design System', 'cat' => 'Frontend CSS', 'tech' => 'CSS3 /
Media Queries', 'path' => '../practical 2/index.html'],
    ['num' => 3, 'name' => 'Interactive DOM Event Engine', 'cat' => 'Client Scripting', 'tech' =>
'JavaScript / Event Listeners', 'path' => '../Practical 3/index.html'],
    ['num' => 4, 'name' => 'Registration Form Validation', 'cat' => 'Client Validation', 'tech' =>
'Regex / JS Validation', 'path' => '../Practical 4/index.html'],
    ['num' => 5, 'name' => 'Dynamic Course Registration', 'cat' => 'DOM Manipulation', 'tech' =>
'JS DOM / LocalStorage', 'path' => '../Practical 5/index.html'],
    ['num' => 6, 'name' => 'PHP Form Processing & Sticky State', 'cat' => 'Backend Core', 'tech' =
> 'PHP $_POST / Sticky Forms', 'path' => '../Practical 6/index.php'],
    ['num' => 7, 'name' => 'JSON & CSV Data Persistence', 'cat' => 'File Handling', 'tech' => 'PHP
Filesystem / CSV Export', 'path' => '../Practical 7/index.php'],
    ['num' => 8, 'name' => 'Cookie Management & Preferences', 'cat' => 'State Tracking', 'tech' =>
'PHP Cookies / setcookie()', 'path' => '../Practical 8/index.php'],
    ['num' => 9, 'name' => 'User Registration & BCRYPT Hashing', 'cat' => 'Authentication', 'tech'
=> 'password_hash() / MySQLi', 'path' => '../Practical 9/index.php'],
    ['num' => 10, 'name' => 'Session Security & RBAC Routing', 'cat' => 'Session Management', 'tec
h' => 'session_regenerate_id() / RBAC', 'path' => '../Practical 10/login.php'],
    ['num' => 11, 'name' => 'Student Management MySQLi CRUD', 'cat' => 'Database CRUD', 'tech' =>
'MySQLi / Search & Filter', 'path' => '../Practical 11/index.php'],
    ['num' => 12, 'name' => 'Event CRUD & Secure Poster Upload', 'cat' => 'File Upload CRUD', 'tec
h' => 'Multi-Tier Validation / .htaccess', 'path' => '../Practical 12/index.php'],
    ['num' => 13, 'name' => 'RESTful JSON API & Fetch CRUD', 'cat' => 'REST API / SPA', 'tech' =>
'PHP JSON API / async fetch()', 'path' => '../Practical 13/index.html'],
    ['num' => 14, 'name' => 'Admin Dashboard & System Audit Log', 'cat' => 'Admin & Auditing', 'te
ch' => 'SQL Aggregation / audit_logs', 'path' => '../Practical 14/index.html'],
    ['num' => 15, 'name' => 'Master Portal & Virtual Host Deployment', 'cat' => 'Full-Stack Integr
ation', 'tech' => 'Apache VHost / Test Suite', 'path' => './index.php']
];
?>
<!
```

OUTPUT & SCREENSHOTS:

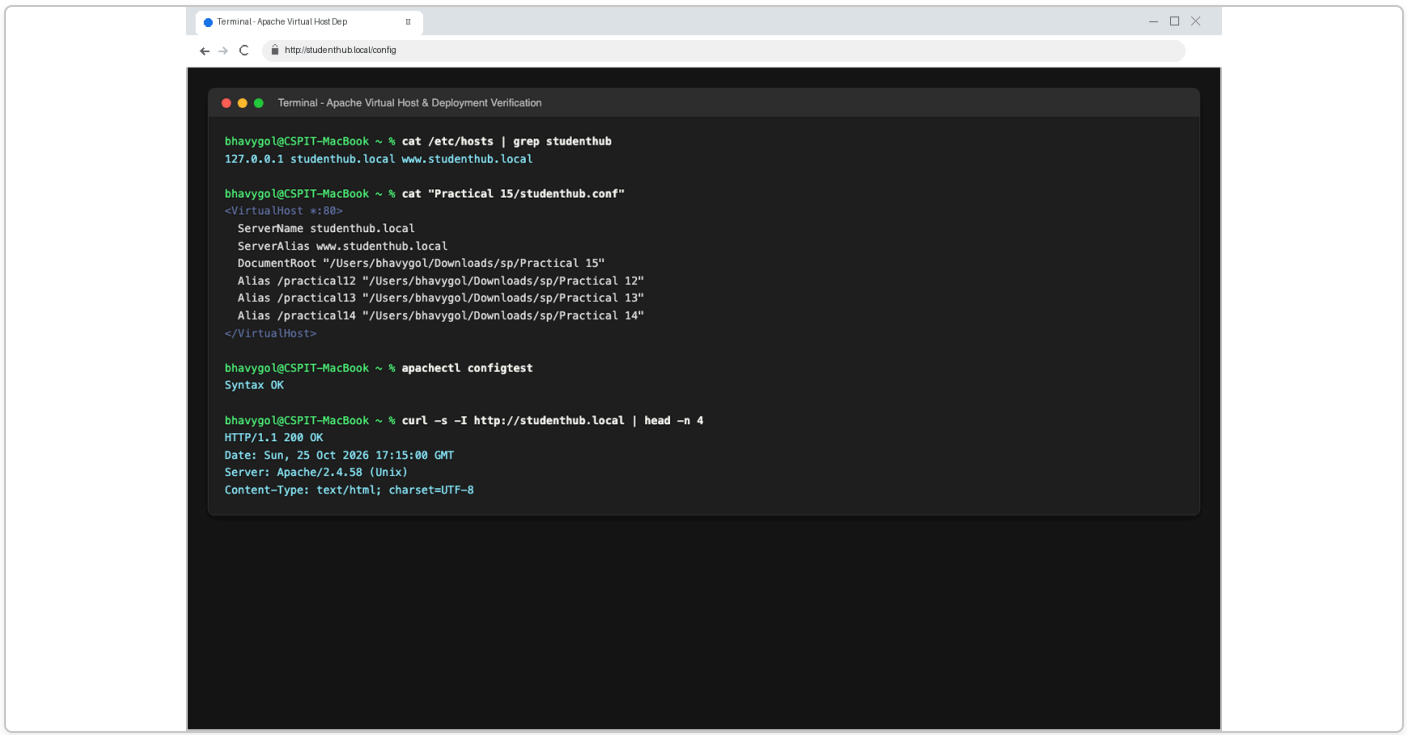
A. Test Case 1 – StudentHub Master Integrated Portal Home & Navigation Dashboard



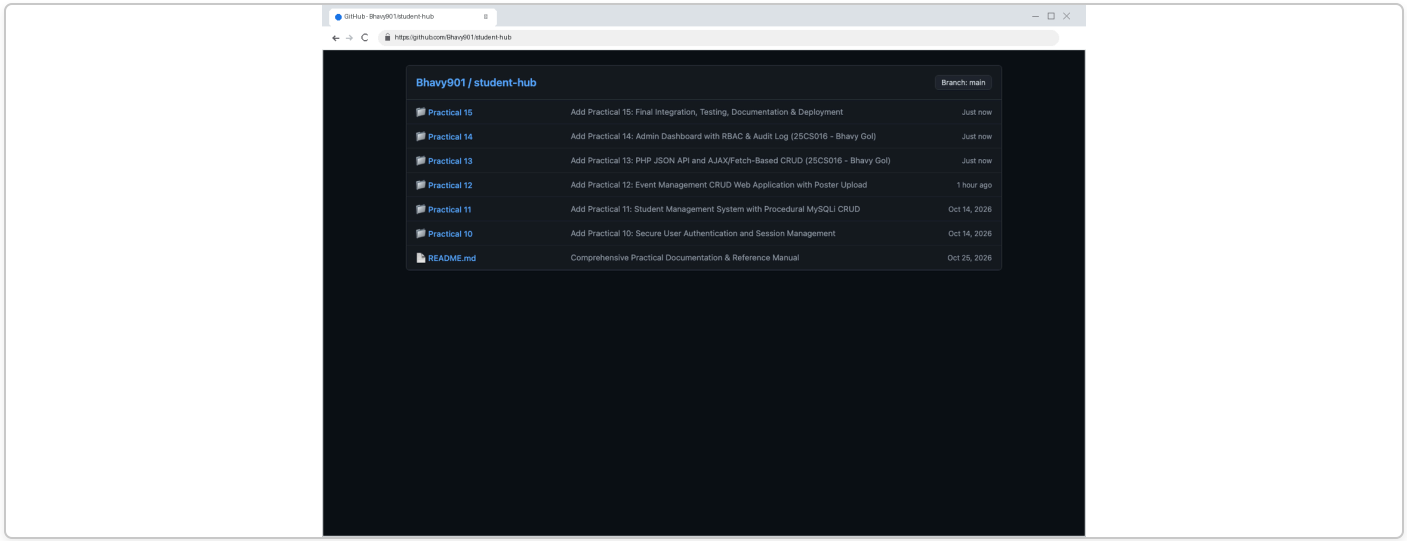
B. Test Case 2 – Automated Integration & Security Test Suite Execution (7/7 Passed)



C. Test Case 3 – Apache Virtual Host Deployment & Local Domain Verification (studenthub.local)



D. Test Case 4 – GitHub Repository & Commit Tree Verification (Bhavy901/student-hub)



Github link:
https://github.com/Bhavy901/student-hub

Learning Outcome

- Successfully synthesized 15 distinct web development modules into a unified, coherent portal.
- Designed programmatic test suites to validate encryption, session lifecycle, and payload integrity.
- Configured Apache Virtual Hosts with domain aliases, directory permissions, and logging.
- Maintained comprehensive academic and technical documentation synchronized with GitHub.